

# Chandra Prakash

✉ chandra.pp@alumni.iitg.ac.in | Guwahati, India |  LinkedIn | Website | GitHub

## RESEARCH INTERESTS

I study quantum fields in nontrivial gravitational backgrounds, ranging from cosmological spacetimes to black-hole geometries. I am particularly interested in **Quantum Field Theory in Curved Spacetime, EFT of Gravity, Asymptotic Symmetries & Holography** and **Early Universe Cosmology**.

## EDUCATION

**Indian Institute of Technology Guwahati (IITG)**, Assam, India

- M.Sc. Physics (GPA: 8.65/10) July 2023 – June 2025
  - Thesis: *From Quantum Fields to Cosmic Structures: Tracing the Early Universe through Correlation Functions*
  - Supervisor: Prof. Dr. Debaprasad Maity
  - Focus: Particle production during inflation, correlation functions, cosmological observables.

**Savitribai Phule Pune University**, Maharashtra, India

- B.Sc. Physics (GPA: 9.38/10) June 2020 – June 2023
  - Thesis: *Non Commutative Geometry as quantization of spacetime*
  - Supervisor: Prof. Lalita Rane
  - Graduated among top 5% of class.

## PUBLICATIONS & PREPRINTS

- **Chandra Prakash** and Rajesh Karmakar. *Canonical quantization of a massive vector field in a Schwarzschild black hole background*. Published in **Physical Review D. (PRD)**, 2026. [arXiv:2605.27047]
- **Chandra Prakash** and Debaprasad Maity. *Evolution of Cosmological Correlators Across Reheating*. Under Review in **Physical Review D. (PRD)**, 2026. [arXiv:2603.24635]
- Anchita Chetia, Nirmalya Kajuri, and **Chandra Prakash**. *Momentum Space Correlation Functions in 2D Galilean Conformal Algebra*. Published in **Journal of High Energy Physics (JHEP)**, 2025. [arXiv:2512.21733]

## RESEARCH EXPERIENCE

**Shanghai University**, China

- Dirac Bracket and QFT in curved spacetime 2026 January – 2026 June
  - **Collaborator:** Dr. Rajesh Karmakar
  - **Research Focus:** Quantum field theory in curved spacetime; black hole backgrounds; vacuum structure.
  - **Skills:** Quantum field theory in curved spacetime; hidden symmetries; black hole physics, constrained hamiltonian dynamics.
  - **Methodology:** Using Analytical methods to study field dynamics in curved spacetime, including basis construction via Löwdin orthogonalization and constrained systems analysis using Dirac-type methods.

**Indian Institute of Technology Mandi**, Himachal Pradesh, India

- Non-relativistic limit of AdS/CFT in momentum space Summer 2025 – 2025 December
  - **Guide:** Dr. Nirmalya Kajuri
  - **Research Focus:** Studying Galilean Conformal algebra in momentum space to deduce the constraints on relativistic AdS/CFT limits.
  - **Skills:** Bootstrap techniques, conformal ward identities, Wigner-Innou contraction.
  - **Methodology:** Analyzed Galilean Conformal Algebra (GCA) via Wigner-Inönü contraction; derived generators in position and momentum space. Deduced and solved Ward identities for two- and three-point functions.

**Indian Institute of Technology Guwahati**, Assam, India

- Cosmological Correlations during Reheating 2024 July – 2025 April
  - **Guide:** Prof. Dr. Debaprasad Maity
  - **Research Focus:** Cosmological collider physics and early-universe particle production.
  - **Skills:** Quantum Field Theory in curved spacetime, Schwinger-Keyldish formalism, cosmological perturbation theory, numerical methods, Python.
  - **Methodology:** Worked on evolving cosmological correlators from the end of inflation to CMB Sky, particle production, and CMB effects.

**TECHNICAL  
EXPERTISE**

**Physics:** Quantum Field Theory in Curved Spacetime; Cosmological Perturbation Theory; Effective Field Theory and Schwinger-Keldish formalism; Constrained Hamiltonian Systems; Black Hole Quantum Physics; Conformal Symmetry and Ward Identities.

**Computational:** Python, C++, Fortran, Mathematica.

**CONFERENCES &  
SCHOOLS**

- Loop the Loop Workshop (2024): Self-force and loop corrections in GR.
- GIAN Course on Standard Model EFT (2024).
- IUCSS Summer School (2021): Lorentz/CPT violation.
- String 2021 Conference.

**OUTREACH**

- Gave a Zoom talk on arXiv:2603.24635 at Swansea Cosmology Group.
- Maintains detailed **Conformal Field Theory**, **String Theory** and **Cosmology** notes at [lunar-photon.github.io](https://lunar-photon.github.io).
- Presented Black Hole Entropy work at Zonal Level Aavishkar research event.
- First prize in poster competition (Pune University).

**REFERENCES**

- **Dr. Nirmalya Kajuri**  
Assistant Professor, School of Physical Sciences  
Indian Institute of Technology Mandi  
Email: [nirmalya@iitmandi.ac.in](mailto:nirmalya@iitmandi.ac.in)
- **Prof. Debaprasad Maity**  
Professor, Department of Physics  
Indian Institute of Technology Guwahati  
Email: [debu@iitg.ac.in](mailto:debu@iitg.ac.in)
- **Prof. Ashwini Kumar Sharma**  
Professor, Department of Physics  
Indian Institute of Technology Guwahati  
Email: [aksharma@iitg.ac.in](mailto:aksharma@iitg.ac.in)